```
SQL> create table dept(dno integer primary key,dname
varchar(30),mgrssn integer,mgr_start date);

Table created.

SQL> insert into dept values('1','cs','11','1-dec-2022');

1 row created.

SQL> insert into dept values('2','account','12','15-
jan-2023');

1 row created.

SQL> insert into dept values('3','it','12','1-jun-2021');

1 row created.

SQL> insert into dept values('4','java','13','1-oct-2023');

1 row created.

SQL> insert into dept values('5','dbms','14','1-nov-2023');

1 row created.

SQL> select * from dept;

       DNO DNAME                              MGRSSN MGR_START
---------- ------------------------------ ---------- ---------
         1 cs                                     11   01-DEC-22
         2 account                                12   15-JAN-23
         3 it                                     12   01-JUN-21
         4 java                                   13   01-OCT-23
         5 dbms                                   14   01-NOV-23


SQL> create table emp23(ssn integer primary key,name
varchar(30),addr varchar(30),gender varchar(10),salary
integer,superssn integer,dno integer,foreign key(dno)
references dept22(dno));

Table created.

SQL> insert into emp
values('11','ibrahim','hubli','m','20000','11','1');

1 row created.

SQL>insert into emp
values('12','umar','hangal','m','25000','11','1');

1 row created.

SQL> insert into emp
values('13','shanwaz','dharwad','m','30000','12','2');
```

```
1 row created.

SQL> insert into emp
values('14','husain','hubli','m','35000','13','2');

1 row created.

SQL>insert into emp
values('15','maaz','haveri','m','40000','14','4');

1 row created.

SQL> select * from emp;

 SSN  NAME      ADDR    GENDER     SALARY   SUPERSSN      DNO
----- ----      ------- ---------- ------ -------------- ----
 11   ibrahim   hubli     m        20000        11         1
 12   umar      hangal    m        25000        11         1
 13   shanwaz   dharwad   m        30000        12         2
 14   husain    hubli     m        35000        13         2
 15   maaz      haveri    m        40000        14         4


SQL> create table delocation(dno integer ,foreign key(dno)
references dept22(dno),dloc varchar(30));

Table created.

SQL> insert into delocation values('1','dharwad');

1 row created.

SQL> insert into delocation values('2','dehli');

1 row created.

SQL> insert into delocation values('3','india');

1 row created.

SQL> insert into delocation values('4','ajmer');

1 row created.

SQL> insert into delocation values('5','agra');

1 row created.

SQL> select * from delocation;

       DNO DLOC
---------- ------------------------------
         1 dharwad
         2 dehli
         3 india
         4 ajmer
```

```
         5 agra

SQL> create table project(pno integer primary key,pname
varchar(30),plocation varchar(20),dno integer ,foreign
key(dno) references dept22(dno));

Table created.

SQL> insert into project values('6','IOT','mumbai','1');

1 row created.

SQL> insert into project values('7','shiru','pune','2');

1 row created.

SQL> insert into project values('8','begin','hangal','3');

1 row created.

SQL> insert into project values('10','starrr','dharwad','5');

1 row created.

SQL> insert into project values('9','waiting','hubli','4');

1 row created.

SQL> select * from project;

     PNO   PNAME              PLOCATION              DNO
---------- ----------------   -----------------     ------
       6   IOT                mumbai                  1
       7   WEB SITE           pune                    2
       8   begin              hangal                  3
      10   starrr             dharwad                 5
       9   waiting            hubli                   4


SQL>  create table work1(ssn integer,foreign key(ssn)
references emp23(ssn),pno integer, foreign key(pno) references
project(pno),hours varchar(10));

Table created.

SQL> insert into work1 values('11','6','8');

1 row created.

SQL> insert into work1 values('11','6','12');

1 row created.

SQL> insert into work values('13','7','9');

1 row created.
```

```
SQL> insert into work values('12','8','10');

1 row created.

SQL> insert into work values('14','9','8');

1 row created.

SQL> select * from work1;

       SSN        PNO HOURS
---------- ---------- ----------
        11          6 8
        11          6 8
        11          6 6
        13          7 9
        12          8 10
        14          9 8
```

i. Make a list of all project numbers for projects that involve an employee whose last name is 'Scott' , either as a worker or as a manager of the department that controls the project.

```
SQL> (select distinct p.pno from project p,dept22 d,emp23 e
where p.dno=d.dno and d.mgrssn=e.ssn and e.name='umar') union
(select distinct p1.pno from project p1,emp23 e1,work w where
p1.pno=w.pno and e1.ssn=w.pno and e1.name='umar');

       PNO
----------
         7
         8
```

ii. Show the resulting salaries if every employee working on the 'IoT' project is given a 10 percent raise.

```
SQL> select e.name,1.1*e.salary as salary from emp23 e,work1
w,project p where e.ssn=w.ssn and w.pno=p.pno and
p.pname='IOT';

NAME                                SALARY
------------------------------ ----------
ibrahim                              22000
ibrahim                              22000
```

iii. Find the sum of the salaries of all employees of the 'Accounts' department, as well as the maximum salary, the minimum salary, and the average salary in this department.

```
SQL> select
sum(e.salary),max(e.salary),min(e.salary),avg(e.salary) from
emp23 e,dept22 d where e.dno=d.dno and dname='account';

SUM(E.SALARY) MAX(E.SALARY) MIN(E.SALARY) AVG(E.SALARY)
------------- ------------- ------------- -------------
   65000           35000         30000         32500
```

iv. Create a view with columns dept name and dept location. Display name of dept located in 'Dharwad' on this view.

```
SQL> create view dlocation as select dname,dloc from dept22
d,delocation l where d.dno=l.dno and dloc='dharwad';

View created.

SQL> select * from dlocation;

DNAME                          DLOC
------------------------------ ------------------------------
cs                             dharwad
```